

In Manohar & Stewart the $\mathcal{O}(\alpha_s)$ NRQCD $Q\bar{Q}$ potential arises from a

v -expansion of the tree-level Coulomb exchange diagram:

$$\begin{array}{c} \vec{p} \longrightarrow \longrightarrow \vec{q} \\ | \text{wavy line} | \\ -\vec{p} \longleftarrow \longleftarrow -\vec{q} \end{array} = \bar{u}(\vec{q})(ig\gamma^\mu t^a)u(\vec{p})\bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)v(-\vec{q}) \frac{-i}{-(\vec{q}-\vec{p})^2} \xrightarrow{\text{v expansion}} \sum_{n=-2}^{\infty} V^{(n,1)}(\vec{p}, \vec{q})$$

where $V^{(n,j)} \sim v^n \alpha_s^j$.

Then some of the $\mathcal{O}(\alpha_s^2)$ NRQCD diagrams can be written in terms of this potential:

$$\int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{\vec{\ell}^2 - \vec{p}^2} \left[\sum_{n=-2}^{\infty} V^{(n,1)}(\vec{p}, \vec{\ell}) \right] \left[\sum_{n=-2}^{\infty} V^{(n,1)}(\vec{\ell}, \vec{q}) \right] \rightarrow \text{this comes from evaluating the residue in the NR quark propagators.}$$

$$\approx \int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{\vec{\ell}^2 - \vec{p}^2} \left[\underbrace{V^{(-2,1)}(\vec{p}, \vec{\ell}) V^{(-2,1)}(\vec{\ell}, \vec{q})}_{1/v^3: \text{Eqn 24 of M\&S}} + \underbrace{V^{(-2,1)}(\vec{p}, \vec{\ell}) V^{(0,1)}(\vec{\ell}, \vec{q}) + V^{(0,1)}(\vec{p}, \vec{\ell}) V^{(-2,1)}(\vec{\ell}, \vec{q})}_{1/v: \text{Eqn 28 of M\&S}} + (\text{higher-order in } v) \right]$$

Note that in order to get these v scalings for these terms, we need $\tilde{l} \sim \bar{p}$, $\tilde{q} \sim v$. That will be important when we relate this to the threshold expansion because that $\tilde{l}$ scaling corresponds to the potential & soft regions.

Now let's look at the full NRQCD expression and write it in terms of full QCD objects:

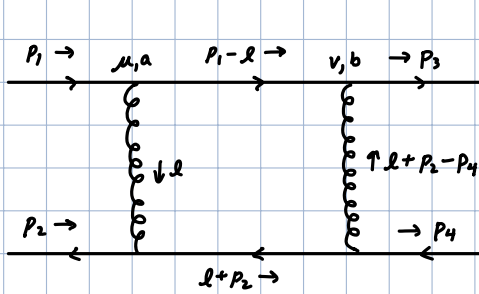
$$\begin{aligned}
& \int \frac{d^3 \vec{x}}{(2\pi)^3} \frac{1}{\vec{x}^2 - \vec{p}^2} \left[\sum_{n=-2}^{\infty} V^{(n,1)}(\vec{p}, \vec{x}) \right] \left[\sum_{n=-2}^{\infty} V^{(n,1)}(\vec{x}, \vec{q}) \right] \\
&= \int \frac{d^3 \vec{x}}{(2\pi)^3} \frac{1}{\vec{x}^2 - \vec{p}^2} \bar{u}(\vec{x})(ig\gamma^\mu t^a)u(p) \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)v(-\vec{x}) \frac{-i}{-(\vec{x}-\vec{p})^2} \bar{u}(\vec{q})(ig\gamma^\nu t^b)u(\vec{x}) \bar{v}(-\vec{x})(ig\gamma_\nu \bar{t}^b)v(-\vec{q}) \frac{-i}{-(\vec{q}-\vec{x})^2} \\
&= \int \frac{d^3 \vec{x}}{(2\pi)^3} \frac{1}{\vec{x}^2 - \vec{p}^2} \frac{-i}{(\vec{x}-\vec{p})^2} \frac{-i}{(\vec{x}-\vec{q})^2} \bar{u}(\vec{q})(ig\gamma^\nu t^b)u(\vec{x}) \bar{u}(\vec{x})(ig\gamma^\mu t^a)u(p) \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)v(-\vec{x}) \bar{v}(-\vec{x})(ig\gamma_\nu \bar{t}^b)v(-\vec{q})
\end{aligned}$$

Then if we sum over the spins for the particle and antiparticle we get

↳ can I do that? Seems like they should have different spin indices

$$= \int \frac{d^3 \vec{q}}{(2\pi)^3} \frac{1}{\vec{q}^2 - \vec{p}^2} \frac{-i}{(\vec{q} - \vec{p})^2} \frac{-i}{(\vec{q} - \vec{q})^2} \bar{u}(\vec{q})(ig\gamma^\nu \vec{t}^b)(\not{x} + m)(ig\gamma^\mu \vec{t}^a)u(p) \bar{v}(-\vec{p})(ig\gamma_\mu \vec{t}^a)(-\not{x} - m)(ig\gamma_\nu \vec{t}^b)v(-\vec{q})$$

This nearly looks like the QCD box diagram with each of the propagators expanded to LO in the potential region. The full expression is



$$\begin{aligned} &\equiv k \sim m \quad \uparrow \\ &\equiv \pm p \sim mv \quad \uparrow \\ &\downarrow \\ &\equiv \pm q \sim mv \end{aligned}$$

$$p_{1/2} = \left(\frac{E}{2}, \pm \vec{p} \right) = \left(\frac{E}{2}, \vec{0} \right) + (0, \pm \vec{p})$$

$$p_{3/4} = \left(\frac{E}{2}, \pm \vec{q} \right) = \left(\frac{E}{2}, \vec{0} \right) + (0, \pm \vec{q})$$

$$\begin{aligned} = & \int \frac{d^4 l}{(2\pi)^4} \bar{u}(p_3) (ig \gamma^{\nu t^b}) \frac{i(p_1 - l + m)}{(p_1 - l)^2 - m^2} (ig \gamma^{\mu t^a}) u(p_1) \bar{v}(p_2) (ig \gamma_{\mu} \bar{t}^a) \frac{i(-l - p_2 + m)}{(l + p_2)^2 - m^2} (ig \gamma_{\nu} \bar{t}^b) v(p_4) \\ & \times \frac{-i}{l^2 - \lambda^2} \frac{-i}{(l + p_2 - p_4)^2 - \lambda^2} \quad (+ie \text{ implicit}) \end{aligned}$$

When the loop momentum is potential, $l_0^2 \sim v^4$ and everything else in the propagators is $\sim v^2$. $\rightarrow$ except λ^2 but that is zero. Expand in l_0^2 . The LO term in each propagator is $\frac{1}{v^2}$.

$$\frac{1}{l^2 - 2l \cdot p \pm 2l \cdot k} = \frac{1}{-\vec{l}^2 - 2l \cdot p \pm 2l \cdot k} + \mathcal{O}(v^0)$$

$$\frac{1}{l^2 - \lambda^2} = \frac{1}{-\vec{l}^2 - \lambda^2} + \mathcal{O}(v^0)$$

$$\frac{1}{(l + q - p)^2 - \lambda^2} = \frac{1}{-(\vec{l} + \vec{q} - \vec{p})^2 - \lambda^2} + \mathcal{O}(v^0)$$

Only the quark propagators have l_0 dependence at LO. The residue gives $\frac{1}{2E} \frac{1}{\vec{l}^2 - 2\vec{l} \cdot \vec{p}}$.

Drop the l_0 dependence of the numerator (since this is the smallest of $l + p + m$) and we get

$$\begin{aligned} = & i \int \frac{d^3 \vec{l}}{(2\pi)^3} \frac{1}{2E} \frac{(i)^2}{\vec{l}^2 - 2\vec{l} \cdot \vec{p}} \frac{-i}{-\vec{l}^2 - \lambda^2} \frac{-i}{-(\vec{l} + \vec{q} - \vec{p})^2 - \lambda^2} \bar{u}(\vec{q}) (ig \gamma^{\nu t^b}) (\not{p}_1 + \vec{\gamma} \cdot \vec{l} + m) (ig \gamma^{\mu t^a}) u(p) \\ & \times \bar{v}(-\vec{p}) (ig \gamma_{\mu} \bar{t}^a) (\vec{\gamma} \cdot \vec{l} - \not{p}_2 + m) (ig \gamma_{\nu} \bar{t}^b) v(-\vec{q}) \end{aligned}$$

Flip $\vec{l} \rightarrow -\vec{l}$ and then shift $\vec{l} \rightarrow \vec{l} - \vec{p}$. Drop the λ dependence.

$$\begin{aligned} = & i \int \frac{d^3 \vec{l}}{(2\pi)^3} \frac{1}{2E} \frac{(i)^2}{\vec{l}^2 - \vec{p}^2} \frac{-i}{-(\vec{l} - \vec{p})^2} \frac{-i}{-(\vec{l} - \vec{q})^2} \bar{u}(\vec{q}) (ig \gamma^{\nu t^b}) (\not{k}_0 \gamma^0 - \vec{\gamma} \cdot \vec{l} + m) (ig \gamma^{\mu t^a}) u(\vec{p}) \\ & \times \bar{v}(-\vec{p}) (ig \gamma_{\mu} \bar{t}^a) (-\vec{\gamma} \cdot \vec{l} - \not{k}_0 \gamma^0 + m) (ig \gamma_{\nu} \bar{t}^b) v(-\vec{q}) \end{aligned}$$

Now compare this to the NRQCD diagrams arising from the contact potential:

$$(\text{QCD box})_p = i \int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{2E} \frac{(i)^2}{\vec{\ell}^2 - \vec{p}^2} \frac{-i}{(\vec{\ell} - \vec{p})^2} \frac{-i}{(\vec{\ell} - \vec{q})^2} \bar{u}(\vec{q})(ig\gamma^\nu t^b)(k_0 \gamma^0 - \vec{\tau} \cdot \vec{\ell} + m)(ig\gamma^\mu t^a)u(\vec{p})$$

$$\times \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)(-\vec{\tau} \cdot \vec{\ell} - k_0 \gamma_0 + m)(ig\gamma_\nu \bar{t}^b)v(-\vec{q})$$

$$(\text{NRQCD}) = \int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{\vec{\ell}^2 - \vec{p}^2} \frac{-i}{(\vec{\ell} - \vec{p})^2} \frac{-i}{(\vec{\ell} - \vec{q})^2} \bar{u}(\vec{q})(ig\gamma^\nu t^b)(\vec{\ell} + m)(ig\gamma^\mu t^a)u(p) \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)(-\vec{\ell} - m)(ig\gamma_\nu \bar{t}^b)v(-\vec{q})$$

Simplify and drop the k_0 in the NRQCD result that probably shouldn't have been there.

$$(\text{QCD box}) = i \int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{2E} \frac{1}{\vec{\ell}^2 - \vec{p}^2} \frac{1}{(\vec{\ell} - \vec{p})^2} \frac{1}{(\vec{\ell} - \vec{q})^2} \bar{u}(\vec{q})(ig\gamma^\nu t^b)(k_0 \gamma^0 - \vec{\tau} \cdot \vec{\ell} + m)(ig\gamma^\mu t^a)u(\vec{p})$$

$$\times \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)(-\vec{\tau} \cdot \vec{\ell} - k_0 \gamma_0 + m)(ig\gamma_\nu \bar{t}^b)v(-\vec{q})$$

$$(\text{NRQCD}) = \int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{\vec{\ell}^2 - \vec{p}^2} \frac{1}{(\vec{\ell} - \vec{p})^2} \frac{1}{(\vec{\ell} - \vec{q})^2} \bar{u}(\vec{q})(ig\gamma^\nu t^b)(-\vec{\tau} \cdot \vec{\ell} + m)(ig\gamma^\mu t^a)u(p)$$

$$\times \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)(-\vec{\tau} \cdot \vec{\ell} + m)(ig\gamma_\nu \bar{t}^b)v(-\vec{q})$$

So we can see that

$$(\text{QCD box})_p = \frac{i}{2E} (\text{NRQCD})$$

→ this might be the $\frac{1}{m}$ out front in Eqs 28 of M&S.

$$+ i \int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{2E} \frac{1}{\vec{\ell}^2 - \vec{p}^2} \frac{1}{(\vec{\ell} - \vec{p})^2} \frac{1}{(\vec{\ell} - \vec{q})^2}$$

$$\times \left[\bar{u}(\vec{q})(ig\gamma^\nu t^b)(k_0 \gamma^0)(ig\gamma^\mu t^a)u(\vec{p}) \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)(-\vec{\tau} \cdot \vec{\ell} + m)(ig\gamma_\nu \bar{t}^b)v(-\vec{q}) \right.$$

$$+ \bar{u}(\vec{q})(ig\gamma^\nu t^b)(-\vec{\tau} \cdot \vec{\ell} + m)(ig\gamma^\mu t^a)u(\vec{p}) \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)(-k_0 \gamma_0)(ig\gamma_\nu \bar{t}^b)v(-\vec{q})$$

$$\left. + \bar{u}(\vec{q})(ig\gamma^\nu t^b)(k_0 \gamma^0)(ig\gamma^\mu t^a)u(\vec{p}) \bar{v}(-\vec{p})(ig\gamma_\mu \bar{t}^a)(-k_0 \gamma_0)(ig\gamma_\nu \bar{t}^b)v(-\vec{q}) \right]$$

Some of these might be zero

Diagrammatically and more compactly, this looks like

$$\left(\text{Diagram: two horizontal lines with two vertical wavy lines between them} \right)_p = \frac{i}{2E} \text{Diagram: two vertices connected by a loop} + \frac{i}{4} g^4 t^a t^a \otimes \bar{t}^a \bar{t}^a \int \frac{d^3 \vec{\ell}}{(2\pi)^3} \frac{1}{\vec{\ell}^2 - \vec{p}^2} \frac{1}{(\vec{\ell} - \vec{p})^2} \frac{1}{(\vec{\ell} - \vec{q})^2}$$

$$\times \left[\gamma^\nu \gamma^0 \gamma^\mu \otimes \gamma_\mu (-\vec{\tau} \cdot \vec{\ell} + m) \gamma_\nu - \gamma^\nu (-\vec{\tau} \cdot \vec{\ell} + m) \gamma^\mu \otimes \gamma_\mu \gamma_0 \gamma_\nu - \frac{\epsilon}{2} \gamma^\nu \gamma^0 \gamma^\mu \otimes \gamma_\mu \gamma_0 \gamma_\nu \right]$$

This is promising but it is still not very careful. We are mixing powers of v

by not expanding the spinors.

I would then guess that the NRQCD soft loop comes from the soft region of the QCD box diagram, but I do not know how to write down the NRQCD numerator to check this.

And I would guess that the only other NRQCD diagram comes from

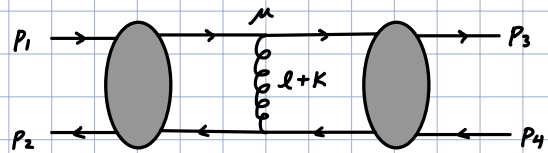
$$\left(\text{Box Diagram} \right)_P \propto \text{V} + (\text{leftover terms})$$

Regarding higher order terms in v ... the NRQCD diagrams get them by just including more terms in the V sum. But in the potential region of QCD you can also get v corrections from higher order terms in the expansion of the propagators, which I cannot see being interpreted as NRQCD diagrams. So this equation only holds for the first few orders in v .

But this is still nice because it means we do not need to evaluate the potential region or the NRQCD diagrams at all to get the matching coefficients.

$$\begin{aligned} & \gamma^\nu \gamma^0 \gamma^\mu \otimes \gamma_\mu (-\vec{\gamma} \cdot \vec{l} + m) \gamma_\nu - \gamma^\nu (-\vec{\gamma} \cdot \vec{l} + m) \gamma^\mu \otimes \gamma_\mu \gamma_0 \gamma_\nu - \frac{\epsilon}{2} \gamma^\nu \gamma^0 \gamma^\mu \otimes \gamma_\mu \gamma_0 \gamma_\nu \\ = & -2 \vec{\gamma} \cdot \vec{l} \otimes \gamma^0 - 2m \gamma^0 \otimes \mathbb{1} - 2 \vec{\gamma} \cdot \vec{l} \gamma^5 \otimes \gamma^0 \gamma_5 - m i \epsilon^{\sigma\nu\alpha\mu} \gamma_\sigma \gamma^5 \otimes \gamma_\mu \gamma_\nu \\ & + 2 \gamma^0 \otimes \vec{\gamma} \cdot \vec{l} + 2m \mathbb{1} \otimes \gamma^0 + 2 \gamma^0 \gamma_5 \otimes \vec{\gamma} \cdot \vec{l} \gamma^5 + m i \epsilon^{\sigma\nu\alpha\mu} \gamma_\mu \gamma_\nu \otimes \gamma_\sigma \gamma^5 - \frac{\epsilon}{2} \gamma^\nu \gamma^0 \gamma^\mu \otimes \gamma_\mu \gamma_0 \gamma_\nu \end{aligned}$$

We really need this to work the other way around though. It seems clear that the Coulomb exchange is what is giving rise to the NRQCD result. Consider a general Coulomb exchange that is part of a $Q\bar{Q} \rightarrow Q\bar{Q}$ loop diagram. (There may be more than one).



$$= P \frac{\bar{u}_3 \Gamma_3 \gamma^\mu \Gamma_1 u_1 \bar{v}_2 \Gamma_2 \gamma_\mu \Gamma_4 v_4}{(l+k)^2 - \lambda^2}$$

The Γ_i depend on an arbitrary number of Dirac indices, and may depend on l . k is some linear combo of the p_i . P is comprised of the denominators of all other propagators, as well as constants.

The next step is to write

$$\Gamma_3 \gamma^\mu \Gamma_1 \otimes \Gamma_2 \gamma_\mu \Gamma_4 = f \gamma^\mu \otimes \gamma_\mu + \Gamma$$

Is there a way to project out what f is?

Write $k = (k_0, \vec{k}) \sim (m, m\mathbf{v})$

$$\frac{1}{(l+k)^2 - \lambda^2} = \frac{1}{l_0^2 - \vec{l}^2 + k_0^2 - \vec{k}^2 + 2l_0 k_0 - 2\vec{l} \cdot \vec{k} - \lambda^2}$$

we have a k_0^2 here that is zero for box diag... how do we deal w/ it in potential region? If it is nonzero then it is bigger than $\vec{l}$, $\vec{k}$.